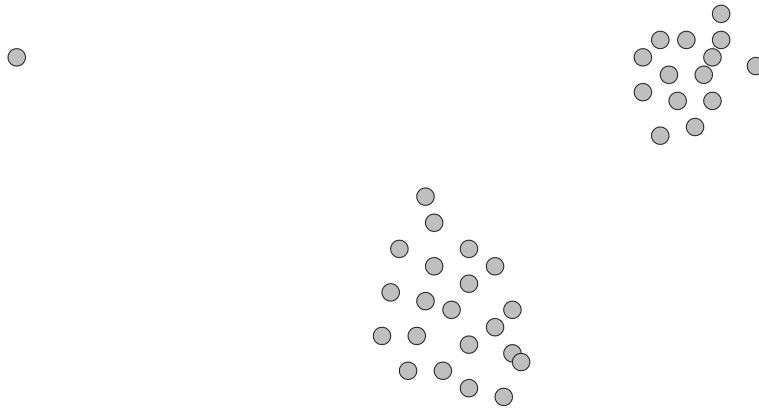


Exercise 1 :

For the following data set:



- identify two initial cluster centers such that 2-means clustering initialized with these cluster centers will terminate with the single outlier forming one cluster, and all other points the second cluster
- identify two initial cluster centers such that 2-means clustering initialized with these cluster centers will terminate with the two big groups of points forming different clusters (and the outlier belonging to one of those)

In both cases indicate the (approximate) position of the final cluster centers.

Exercise 2 :

Consider the data points plotted in Figure 1.

- Perform two iterations of the k -means algorithm using
 - the data points (2, 6) and (3, 5) as the initial cluster centers.
 - the Euclidean distance as distance metric
- Calculate the sum of squared errors using the initial cluster centers and the cluster centers that you found above.

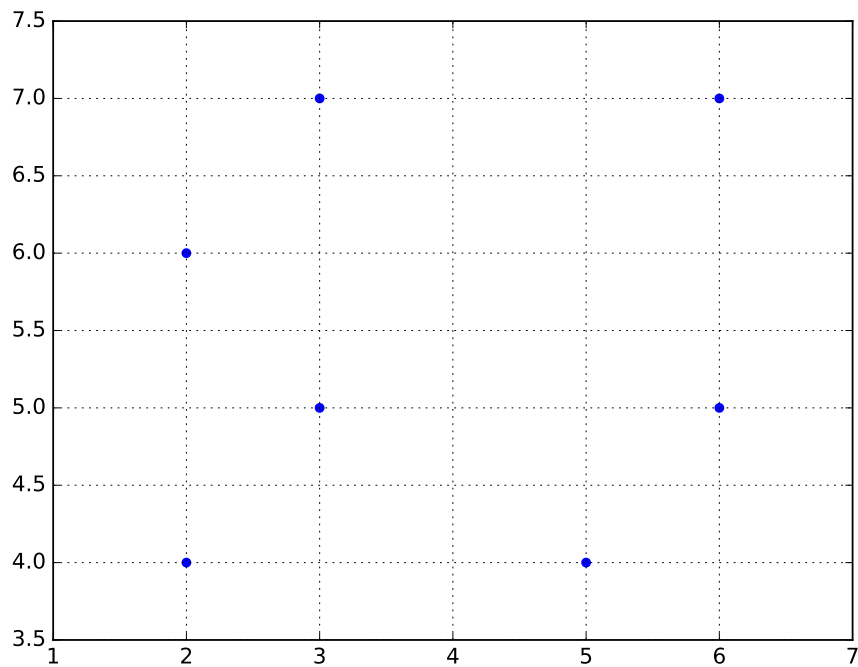


Figure 1: Data to be clustered in Exercise 2.

Exercise 3 :

Perform one more iteration of the EM algorithm for the example on Slide 33. Note that you will first need to complete the last two maximization calculations for the 2nd iteration, which is left unfinished on the slides.

Exercise 4 :

Consider the following five data points living in $\mathbb{R}^2$:

d_1	(7, 2)
d_2	(52, 3)
d_3	(70, 10)
d_4	(85, 11)
d_5	(90, 8)

Using the Euclidean distance to measure distances:

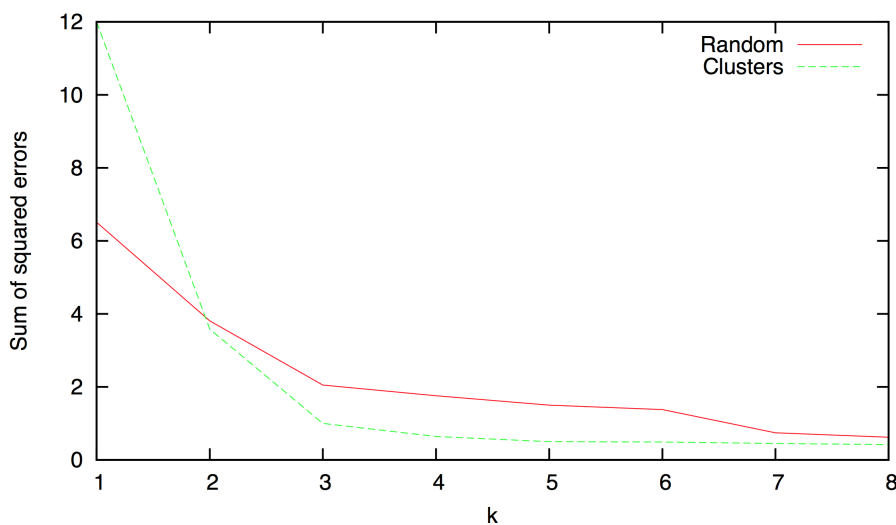
- Find the data point closest to d_2 in the data set, i.e., (52, 3).
- Normalize the data using Z-score normalization. Find the data point closest to d_2 using the transformed data set.

- (iii) Let the data points $(7, 2)$ and $(70, 10)$ be initial cluster centers for the k -means algorithm. Perform one more k -means iteration by updating these cluster centers using the non-normalized data set above.
- (iv) Do you see any potential problems in directly using k -means clustering with the Euclidean distance based the data set above? What could you do to mitigate such problems?

Exercise 5 :

1.

We perform k -means clustering for $k = 1, 2, 3, 4, 5, 6, 7, 8$ on two datasets (Random and Clusters). sets. For each of the cases, we compute the “Within cluster sum of squared errors” (which corresponds to the sum of squared errors, the distance for each example to the centroid of its cluster). How can plots like these be used to determine the “right” number of clusters?



- 2. Suppose that, for a given value of k , we run k -means multiple times with different random seeds. How could we compare the results obtained in the different runs to determine if the dataset has k “real” clusters?